

1 关于规范排版

- 微分符号用正体，区别 dx 和 $\mathrm{d}x$ ；
- 常数用正体，区别 e , i 和 e , i ；
- 导数：

$$y'(x) = \frac{\mathrm{d}y}{\mathrm{d}x}$$

- 角标用正体，区别 Y_{MCMC} 和 Y_{MCMC} ；
- 矩阵和向量 $\mathbf{Ax} = \mathbf{b}$ ；

规范排版的欧拉公式：

$$\mathrm{e}^{\mathrm{i}\theta} = \cos \theta + \mathrm{i} \sin \theta \quad (1)$$

2 关于 SI 包 siunitx

- 写数字：
 - 千分位：123 456.789；
 - 科学计数法： 3.14×10^{-5} ；
- 写单位： $\mathrm{kg} \cdot \mathrm{m/s}$ 和 $\mathrm{J/(mol \cdot K)}$ ；
- SI 指令： $9.8 \mathrm{m/s^2}$ 和 $1.602 \times 10^{-19} \mathrm{C}$ ；
- 公式中：

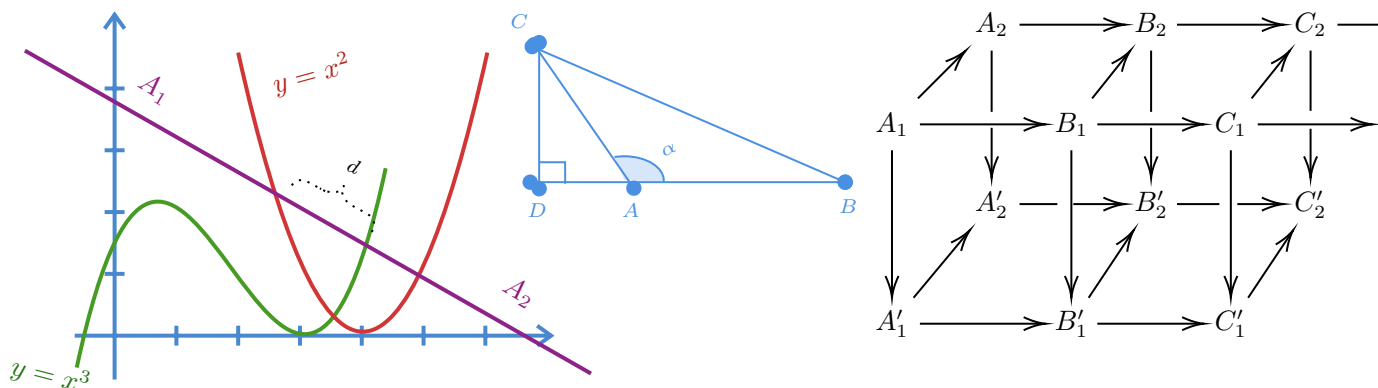
- 不确定度：

$$R = (8.314 \pm 0.001) \mathrm{J/(mol \cdot K)}$$

- 完整公式：

$$E = \frac{1}{2}mv^2 = 0.5 \mathrm{kg} \cdot (10 \mathrm{m/s})^2 = 50 \mathrm{J}$$

3 测试 Tikz



4 V1.3 新增

4.1 测试新增色块

Claim 4.1:

这是盒子 `uclaim` 的内容，文字、公式、图片都可以放在这里。

Wishing your day stays bright —like sunshine that chose to stay just for you —is more than just a kind thought. It's a quiet hope that the world bends a little in your favor today. Imagine the morning sun pausing in its arc across the sky, lingering not out of duty, but out of devotion. Its golden light drapes gently across your shoulders, not with heat, but with warmth —a presence, a reminder that joy doesn't always shout. Sometimes, it simply shines.

Life moves quickly, often tangled in shadows and shifting moods, but today —just today —let the light settle in. Let it follow you from window to window, from task to task, not as a spotlight but as a companion. Let it spill into the cracks of doubt, soften the edges of worry, and warm the corners where dreams wait patiently. And if clouds should gather, as they sometimes do, may the memory of that steady sunshine carry you —a whisper of light in your pocket, proof that brightness once chose you, and could again.

讲述者, narrator

这是盒子 narrator 的内容, 文字、公式、图片都可以放在这里。

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通用 2, ubox2

这是盒子 ubox2 的内容, 文字、公式、图片都可以放在这里。

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4.2 测试颜色效果

- 这是高亮: Wishing your day stays bright ——like sunshine that chose to stay just for you.
- 这是强调: Wishing your day stays bright ——like sunshine that chose to stay just for you.
- 这是标记: Wishing your day stays bright ——like sunshine that chose to stay just for you.
- 这是注记: Wishing your day stays bright ——like sunshine that chose to stay just for you.
- 推荐颜色:
 - 漫光天空, SkyRelax;
 - 优雅的, Elegant;
 - 敦煌, DungHuang;

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附录

A 公式表

$$\Gamma(z)\Gamma(1-z) = \frac{\pi}{\sin(\pi z)}$$

$$\Gamma(2z) = 2^{2z-1} \sqrt{\pi} \Gamma(z) \Gamma\left(z + \frac{1}{2}\right)$$

$$x! \approx \sqrt{2\pi x} \left(\frac{x}{e}\right)^x \quad (\operatorname{Re} x \gg 1)$$

$$J_\nu(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(k + \nu + 1)} \left(\frac{x}{2}\right)^{2k+\nu}$$

$$N_\nu(x) = \lim_{\alpha \rightarrow \nu} \frac{J_\alpha(x) \cos(\alpha\pi) - J_{-\alpha}(x)}{\sin(\alpha\pi)}$$

$$\frac{d}{dx} [x^\nu Z_\nu(x)] = x^\nu Z_{\nu-1}(x)$$

$$\frac{d}{dx} [x^{-\nu} Z_\nu(x)] = -x^{-\nu} Z_{\nu+1}(x)$$

$$Z_{\nu-1}(x) - Z_{\nu+1}(x) = 2Z'_\nu(x)$$

$$Z_{\nu-1}(x) + Z_{\nu+1}(x) = \frac{2\nu}{x} Z_\nu(x)$$

$$J_\nu(x) \approx \sqrt{\frac{2}{\pi x}} \cos\left(x - \frac{\nu\pi}{2} - \frac{\pi}{4}\right) \quad (x \gg \sqrt{|\nu^2 - \frac{1}{4}|})$$

$$N_\nu(x) \approx \sqrt{\frac{2}{\pi x}} \sin\left(x - \frac{\nu\pi}{2} - \frac{\pi}{4}\right) \quad (x \gg \sqrt{|\nu^2 - \frac{1}{4}|})$$

$$J_m(x) = \frac{1}{2\pi} \int_a^{a+2\pi} e^{i(x \sin \theta - m\theta)} d\theta \quad (m \in \mathbb{Z}, a \in \mathbb{R})$$

$$e^{\frac{x}{2}(t-\frac{1}{t})} = \sum_{n=-\infty}^{\infty} J_n(x) t^n$$

$$\int_0^1 x J_m(\mu_i x) J_m(\mu_j x) dx = \frac{\delta_{ij}}{2} [J_{m+1}(\mu_i)]^2 \quad (\mu_i : J_m \text{ 的第 } i \text{ 个正零点})$$

$$\int_0^1 x J_m(\lambda_i x) J_m(\lambda_j x) dx = \frac{\delta_{ij}}{2} \left(1 - \frac{m^2}{\lambda_i^2}\right) [J_m(\lambda_i)]^2 \quad (\lambda_i : J'_m \text{ 的第 } i \text{ 个正零点})$$

$$\int_0^\infty J_m(k_1 r) J_m(k_2 r) r dr = \frac{\delta(k_1 - k_2)}{k_1} \quad (k_1, k_2 > 0)$$

$$j_\ell(x) = \sqrt{\frac{\pi}{2x}} J_{\ell+\frac{1}{2}}(x)$$

$$n_\ell(x) = \sqrt{\frac{\pi}{2x}} N_{\ell+\frac{1}{2}}(x)$$

$$j_\ell(x) = (-x)^\ell \left(\frac{1}{x} \frac{d}{dx} \right)^\ell \left(\frac{\sin x}{x} \right)$$

$$n_\ell(x) = -(-x)^\ell \left(\frac{1}{x} \frac{d}{dx} \right)^\ell \left(\frac{\cos x}{x} \right)$$

$$Y_{\ell m}(\theta, \phi) = \frac{1}{2^\ell \ell!} \sqrt{\frac{(2\ell+1)}{4\pi} \cdot \frac{(\ell-m)!}{(\ell+m)!}} \left[\sin^m \theta \left(\frac{1}{\sin \theta} \frac{d}{d\theta} \right)^{\ell+m} \sin^{2\ell} \theta \right] e^{im\phi}$$

$$P_\ell(\cos \theta) = \sqrt{\frac{4\pi}{2\ell+1}} Y_{\ell 0}(\theta, \phi)$$

$$P_\ell(x) = \sum_{k=0}^{\ell} \frac{(\ell+k)!}{(k!)^2(\ell-k)!} \left(\frac{x-1}{2} \right)^k$$

$$P_\ell(x) = \sum_{k=0}^{\ell} \frac{(-1)^{\ell-k}(\ell+k)!}{(k!)^2(\ell-k)!} \left(\frac{x+1}{2} \right)^k$$

$$P_{2n}(x) = \frac{1}{2^{2n}} \sum_{k=0}^n \frac{(-1)^{n-k} [2(n+k)]!}{(n+k)!(n-k)!(2k)!} x^{2k}$$

$$P_{2n+1}(x) = \frac{1}{2^{2n+1}} \sum_{k=0}^n \frac{(-1)^{n-k} [2(n+k+1)]!}{(n+k+1)!(n-k)!(2k+1)!} x^{2k+1}$$

$$P_\ell(x) = \frac{1}{2^\ell \ell!} \frac{d^\ell}{dx^\ell} (x^2 - 1)^\ell$$

$$(2\ell+1)xP_\ell(x) = (\ell+1)P_{\ell+1}(x) + \ell P_{\ell-1}(x)$$

$$(2\ell+1)P_\ell(x) = P'_{\ell+1}(x) - P'_{\ell-1}(x)$$

$$\int_{-1}^1 P_m(x) P_n(x) dx = \frac{2}{2m+1} \delta_{mn}$$

$$\frac{1}{\sqrt{1+t^2-2xt}} = \sum_{\ell=0}^{\infty} P_\ell(x) t^\ell \quad \text{for } t < |x \pm \sqrt{x^2-1}|$$

$$P_\ell(\mathbf{n}_1 \cdot \mathbf{n}_2) = \frac{4\pi}{2\ell+1} \sum_{m=-\ell}^{\ell} Y_{\ell m}^*(\mathbf{n}_1) Y_{\ell m}(\mathbf{n}_2)$$